

Linear Algebra Theory Project

Objective

This project demonstrates the use of Linear Algebra techniques on student performance data. The aim is to understand vectors, matrices, eigenvalues, decompositions, PCA, and LDA.

Vector Fundamentals

A vector is an ordered list of numbers. In this project, each student's marks are represented as vectors. Norms are used to measure vector length, while dot products measure similarity between vectors.

Matrix Operations

Matrices are rectangular arrays of numbers. Student scores form a matrix where rows represent students and columns represent subjects. Operations include transpose, determinant, and inverse.

Eigenvalues and Eigenvectors

Eigenvalues measure variance direction while eigenvectors show the direction of maximum variance. These are important in PCA and dimensionality reduction.

LU Decomposition

LU decomposition breaks a matrix into Lower and Upper triangular matrices. It helps solve linear equations efficiently.

Singular Value Decomposition (SVD)

SVD decomposes a matrix into U, S, and V matrices. It is widely used in data compression and dimensionality reduction.

Principal Component Analysis (PCA)

PCA reduces dimensions while preserving maximum variance. It converts original variables into principal components.

Linear Discriminant Analysis (LDA)

LDA is used for classification and dimensionality reduction by maximizing class separability.

Conclusion

This project explains practical applications of Linear Algebra concepts in Data Science, Machine Learning, and Artificial Intelligence.